

Using Hierarchical Models to Estimate Heterogeneous Effects

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Abstract

Conditional theories are ubiquitous in political science, but standard interaction models cannot handling this proliferation of theoretically informed modifiers. This paper describes why, when, and how to use hierarchical models to encompass multiple sources of heterogeneity and regularize estimates. Doing so is simple, and starts with identifying the key sources of heterogeneity and grouping observations accordingly. Next, fit a hierarchical model with two linked regressions, one connecting treatment with the outcome, and another that models the treatment effects with potential sources of heterogeneity and partially pools group estimates. This captures systematic and random variation in heterogeneous effects, encompasses many interactions, and fits commonly used modeling frameworks, but regularization in this model breaks down if there are many small groups. I document these claims with a simulation analysis and extension of a published study.

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Social scientists often propose conditional theories (Clark and Golder, 2023) and are interested in how different units respond to the same stimulus for theoretical, normative or policy reasons. But while most theories consider a single modifier, many theories proposing distinct modifiers for the same independent variable suggest that many independent variables have multiple modifiers. For example, international relations experiments on audience costs (Tomz, 2007) have considered how foreign policy dispositions (Kertzer and Brutger, 2016), partisanship (Levendusky and Horowitz, 2012), gender (Barnhart et al., 2020; Schwartz and Blair, 2020) and policy preferences (Chaudoin, 2014) modify reactions to a leader backing down from a threat. Other audience cost experiments manipulate casualties (Davies and Johns, 2013), how the leader backs down (Lin-Greenberg, 2019), other elite reactions (Levendusky and Horowitz, 2012) and promises vs threats (Levy et al., 2015).

Mushrooming modifiers are evident in how scholars conduct applied quantitative work. In the *American Political Science Review*, *American Journal of Political Science* and *Journal of Politics*, more and more researchers estimate some modifier of treatment effects, either to test their argument or explore others. Figure 1 documents the prevalence of heterogeneous effects estimates from interactions or machine learning in Harvard Dataverse replication data from the *American Political Science Review*, *American Journal of Political Science*, and *Journal of Politics* from 2010 to 2024. In this figure I count the number of papers that fit at least one interaction or machine-learning model of effect heterogeneity in their replication data. I did this by downloading all quantitative replication materials hosted in the Harvard dataverse, and then using regular expressions to capture the presence of these techniques. The share of quantitative papers that estimate heterogeneous effects has risen steadily. In 2024, almost 70% of replication files contain at least one heterogeneous effects estimate. These replication files likely understate the extent of conditional theories, as they do not capture arguments with qualitative or game-theoretic evidence.

Unfortunately, existing tools are ill-suited to test this proliferation of conditional arguments

Prevalence of Heterogeneity Techniques

APSR, AJPS and JOP- 2010 to 2024

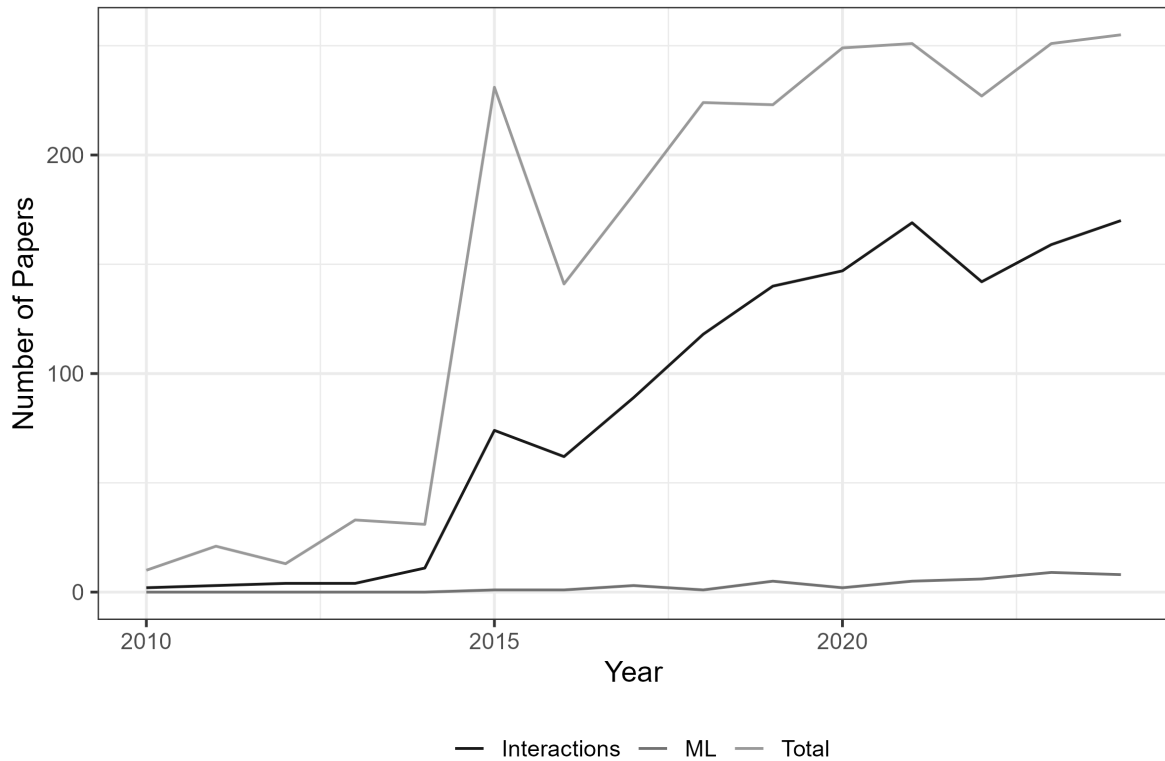


Figure 1. *Count of papers using interactions or machine-learning to estimate heterogeneous effects in three field political science journals from 2010 to 2024. Presence of different methods estimated by parsing replication files hosted by the Harvard Dataverse.*

in social science. Scholars cannot ignore heterogeneity, but the most common tools either increase the risk of spurious results or are hard to interpret and use. Interaction terms and subgroup analysis are ubiquitous because they are relatively easy to interpret, but they increase the risk of spurious results. Many political science analyses have low power even to detect main effects (Arel-Bundock et al., 2025). Adequate power for estimates of even a single interaction can require significantly more data (Gelman, 2018), which may be prohibitively expensive or impossible. As a result, statistically significant heterogeneous effect estimates may be far too large—the result of noise in the data, not systematic differences. This problem is partially responsible for widespread issues replicating findings based on interactions (Simmons, Nelson and Simonsohn, 2011). Adding more modifiers makes interpretation harder and increases these power problems. Researchers might instead fit separate pairwise interactions for each modifier, but multiples models and corresponding testing problems.

Given multiple sources of heterogeneity, machine-learning tools such as random forests (Green and Kern, 2012; Wager and Athey, 2018), support vector machines (Imai and Ratkovic, 2013), and ensemble methods (Grimmer, Messing and Westwood, 2017; Künzel et al., 2019; Dorie et al., 2022) are more likely to avoid over-fitting. These machine learning algorithms usually have some regularization component and can discover complex patterns and high-dimensional variation across multiple modifiers.¹ These tools can be difficult to interpret and implement, however, especially in smaller social science datasets. The relative lack of theoretical guidance and interpretability is especially problematic for testing the relative weight of multiple conditional arguments. If researchers want to compare the impact of different modifiers, machine learning models lack analogues to more familiar regression techniques. This is one reason why these techniques are far less popular, as Figure 1 shows.

¹LASSO models sometimes address similar issues (Ratkovic and Tingley, 2017). Blackwell and Olson (2022) describe a lasso approach to interactions of a single modifier and all the variables in a model that sits between machine learning and linear regressions. This considers a single interaction of interest and bias from other unmodeled interactions with controls.

Despite a large statistics literature documenting how regression-based interpretation and regularization make Bayesian hierarchical models a useful tool for heterogeneous effects estimation (e.g., Feller and Gelman (2015); McElreath (2016); Dorie et al. (2022)), political scientists rarely use this tool. For instance, of the three applied political science citations of Feller and Gelman (2015), only Marquardt (2022) models treatment effects. This paper explains how, when and why to use hierarchical models of heterogeneous effects.

The hierarchical strategy I advocate here addresses the power shortcomings of interactions while allowing more theoretical guidance and comparison than machine learning. Using hierarchical models is more flexible than standard interactions but easier to implement and interpret than machine learning. It preserves a straightforward structure while accommodating more factors and ameliorating the downsides of subgroup analysis. This facilitates argument testing. The main weaknesses are that unlike machine learning, the hierarchical approach lacks the flexibility to discover high-dimensional heterogeneity and regularization will break down if there are many small groups. It also will be slow to fit on large datasets, depending on the underlying sampler.²

There are two key steps when theory and data make using hierarchical models worthwhile. First, researchers should identify potential modifiers of a treatment effect and use them to group observations and model treatment effects. Second, they should take that model of treatment effects and connect it to a model linking the treatment and outcome. Modeling heterogeneous effects in this way produces interpretable results, which facilitates argument testing. It also allows researchers to compare different sources of heterogeneous effects and describe effect variation. These are crucial advantages in a world with many conditional theories.

While frequentist estimation of hierarchical models is possible, Bayesian estimation is easy, more informative and more robust (Stegmueller, 2013). Bayesian estimation provides crucial

²Goplerud (2021) introduces a model that uses Bayesian structured sparsity to estimate which group coefficients are similar and which are different. In this approach, researchers use theory to inform potential groups, but the data determines common estimates for groups.

information by connecting parameters through common prior distributions, thereby regularizing estimates and propagating uncertainty. Working with posterior distributions also gives researchers more flexibility to describe how and when effects vary and use prior predictive checks to examine the assumptions in their models. While computation and coding were once a barrier to employing Bayesian methods, fitting a wide range of hierarchical models is straightforward with the `brms` package in R (Bürkner, 2017).³

In the remainder of this paper, I start by describing how and when to estimate hierarchical models of heterogeneous effects. I then use simulations to compare OLS and hierarchical estimates of treatment effects under different conditions. Finally, I demonstrate the process by analyzing a study of how military alliances shape public support for war by Tomz and Weeks (2021). The reanalysis changes conclusions about the role of high interest in the news and documents the importance of regularization in analyzing subgroups derived from combinations of experimental treatments.

1 Hierarchical Modeling of Heterogeneous Effects

There are two steps in hierarchical models of heterogeneous effects. First, researchers must identify potential sources of heterogeneity, and think about the right model of heterogeneity. This will also depend on what variation is most important and interesting. Theory, policy concerns, or normative factors are all possible motivations.

This first step determines what heterogeneous effects a researcher estimates. It is analogous to researchers thinking through a regression specification and requires the same sort of care. Researchers need to define what variation is most important, link heterogeneous effects to theory, and structure modeling. This is especially important for pre-registration, and could reduce the amount of exploratory analyses in registrations. Not thinking carefully about sources of

³I provide example code below and in the appendix.

heterogeneity will obfuscate results and can hinder model fitting.

There are three general approaches to defining key modifiers. First, researchers can use combinations of other treatments, especially when an intervention has several dimensions but theory emphasizes one of them. The experimental design determines modifiers, and the model estimates heterogeneous treatment effects. If researchers want to know how different issues shape the impact of elite foreign policy cues (Guisinger and Saunders, 2017), they could include indicators of issues, for instance. A similar application of hierarchical estimators for topic-sampling experiments estimates how a treatment effect varies across topics (Clifford and Rainey, 2023).

The most common practice in estimating heterogeneous treatment effects is fully crossed interactions. This estimates the impact of a treatment across experimental strata, but risks spurious results by functionally estimating subgroup results. Most social science papers do not have adequate power for main effects (Arel-Bundock et al., 2025), however, let alone small subgroups that often have 50 or fewer data points.

A second approach uses unit, demographic and contextual factors to estimate effect heterogeneity. Here, researchers examine what factors within or around units shape their response to an independent variable. Researchers could use a mix of individual and contextual factors to predict divergent consequences of a survey experiment treatment. Such a model might include factors such as an indicator of state of residence, age, gender, and race.⁴

For example, Alley (2021) uses alliance characteristics to examine when alliance membership increases or decreases military spending. He models the impact of alliance participation as a function of treaty depth, partner democracy, conditions on military support, issue linkages, democratic membership, foreign policy concessions and other factors. All of these variables are potential sources of credibility or confounding factors. Democratic alliances have higher depth (Martin, 2005), so this model of heterogeneity accounts for potential confounding, and finds

⁴Extrapolations to a representative sample of a subpopulation might require poststratification.

that after accounting for depth, democracy does not impact the relationship between alliances and defense spending, contrary to DiGiuseppe and Poast (2018).

Third, researchers might use hierarchical models to address specific policy concerns. Policy analysts often want to know how an intervention impacts a specific population. Researchers might want to know if a job-training program improves employment outcomes for black women in the South, for instance. To do this, a researcher might specify a heterogeneous effects model with race, gender and region, plus additional controls or other factors.

After defining moderators and how they relate to treatment effects, the second step is fitting a hierarchical model that links a model of the outcome with a model of heterogeneity. Essentially, researchers model the outcome and the process that produces heterogeneous treatments. The model employs two connected regressions. One regression deals with the outcome, and includes the treatment effects.⁵

The other regression models the treatment effects. In this regression, theory should inform which modifiers predict the treatment effect. To regularize the estimates, this second equation must include varying intercepts for groups within the data. Groups can be based on combinations of discrete modifiers and binned continuous modifiers. To maximize variation and preserve regularization, researchers should use continuous modifiers directly in the regression and bin them for setting groups. Not binning modifiers will create many small groups, leading the hierarchical component to break down. As a result, this model includes both systematic and random variation in the impact of a key independent variable on different groups.

Setting groups is a critical task that requires balancing detail against regularization. Larger groups can deviate more from the overall mean, as their data contributes more to the posterior than the prior. A few small groups means less need for a hierarchical model. Smaller groups

⁵If other units such as states define the groups, rather than combinations of modifying variables, then adding group-level predictors to this equation is essential. For example, in a model where an effect varies by state, adding state-level variables like ideology, population and GDP would avoid partially pooling small groups too far towards the overall mean.

will be shrunk more towards the overall mean. Too many small groups and the regularization component will have less information to work with. Using model fit criteria such as WAIC or leave-one-out cross validation can help guide group formulation.

Should the variables in the heterogeneity equation also appear in the outcome equation? Here, the guidance of Brambor, Clark and Golder (2006) to include constituent terms in an interaction will often apply. If a modifier is correlated with treatment and the outcome, then it is also a useful control, as it has an independent relationship with the dependent variable in addition to modifying the treatment effects. The functional form of the two equations may differ, however.

I now briefly describe a generic hierarchical model. For ease of exposition, consider making between-unit comparisons based on an experimental treatment. Start with N units indexed by i , some of which receive a binary treatment T . Assume that the outcome variable y is normally distributed with mean μ_i and standard deviation σ .⁶ g indexes researcher-defined groups based on combinations of the modifying variables.

The outcome for each unit depends on an overall intercept α , an optional matrix of control variables $\mathbf{X}$, and a set of group treatment effects λ_g . When T is binary, estimated λ parameters for untreated units have no impact on the outcome and drop out of the likelihood. For a continuous treatment, effects will depend on the product of T_i and λ .

$$y_i \sim N(\mu_i, \sigma) \quad (\text{Likelihood})$$

$$\mu_i = \alpha_g + \lambda_g T + \mathbf{X}\beta \quad (\text{Outcome Equation})$$

$$\lambda_g = \theta_g + \mathbf{Z}\gamma \quad (\text{Heterogeneity Equation})$$

$$\theta_g \sim N(\mu_\theta, \sigma_\theta) \quad (\text{Group Varying Intercepts})$$

⁶Researchers can and should use binary, categorical and other outcome likelihoods.

The heterogeneity equation then models those group treatment effects with a systematic and random component. The systematic component is a matrix of predictors $\mathbf{Z}$ and associated parameters γ . Z can mirror any regression specification researchers might use for an outcome; linear combinations of variables, interactions, smooth functions. Interactions of the modifiers could capture processes where combinations of modifiers produce non-additive jumps in heterogeneity, for instance.

The random component of the heterogeneity equation is a series of group-specific varying intercepts θ_g . These are critical, because they capture group deviations from the systematic trends expressed in the design matrix $\mathbf{Z}$. Group outliers in the λ estimates are partially pooled back towards the overall mean μ_θ . The variance parameter σ_θ controls the dispersion of the group effects around the overall mean.⁷

The above model can be fit with Bayesian or frequentist methods, but Bayesian estimation offers important advantages. First, it is more flexible, and including prior information can facilitate model fitting. Priors on the α , β , and γ parameters will matter more in very small datasets. The more important prior is for σ_θ , and depending on beliefs about potential dispersion across groups, half-normal or half student-t priors are common choices. This prior is what helps regularize estimates by pulling extreme groups towards the overall mean μ_θ .

Bayesian modeling offers additional information and a wealth of model-checking tools. STAN and brms, the software I provide guidance for here, facilitates prior predictive checks, where researchers simulate from the model to check what estimates are plausible with specific priors. Working with posterior distributions then provides a wealth of information about effect heterogeneity and propagates uncertainty. Last, Bayesian models can provide useful diagnostics such as divergent transitions that can be warnings of specification problems. It is often better to modify a model with fitting problems instead of brute-forcing it through the

⁷In brms, using non-linear syntax can express a model with a treatment, two controls, and three modifiers as: $y \sim \text{lambda} * \text{treat} + \text{controls}$, $\text{lambda} \sim \text{mod1} + \text{mod2} + \text{mod3} + (1 | \text{mod1} + \text{mod2} + \text{mod3})$, $\text{controls} \sim \text{control1} + \text{control2}$, $\text{nl} = \text{TRUE}$

sampler.

Depending on the problem, researchers can extend this initial model in many ways. The generative core of Bayesian hierarchical models adds significant flexibility. For instance, they might model the variance of the treatment effects or the outcome. Fitting splines or smooth functions is also possible, or researchers might extend the model in continuous group space with Gaussian process regression.

In interpreting these estimates, researchers should leverage the full range of information from the different parameters. First, the λ posteriors give the impact of the treatment on each group, and are the core quantity of interest. All λ s reflect a systematic component from the predictors in $\mathbf{Z}\gamma$ and random variation from θ . γ parameters can, depending on the regression, be interpreted as the impact of a change in a modifier on the treatment effects. For example, a γ of .1 on a binary modifier means that λ is .1 higher in expectation when the modifier is one, and .1 lower when it is zero. σ_θ thus measures treatment variation outside the systematic regression. Other techniques such as interactions in OLS with robust standard errors provide less information.

2 When to Use Hierarchical Models

In deciding whether to use a hierarchical model of heterogeneous effects, researchers must weigh specific advantages and disadvantages. In general, estimating heterogeneous effects in this way has three advantages. First, researchers can make detailed inferences about heterogeneous effects in an interpretable framework that encompasses many potential modifiers. This helps examine theories that predict how an effect varies and compare sources of variation.⁸ Partial pooling also facilitates reasonable estimates for small groups by sharing information across groups and incorporating predictors in the heterogeneous effects equation. Finally, this

⁸Rescaling variables in the heterogeneous effects equation can aid model fitting and coefficient comparisons (Gelman, 2008).

approach will be faster than machine learning approaches for many datasets as well as easier to use in small datasets.

Like all methods, the hierarchical approach has downsides, some of which can be ameliorated with modifications, while others should lead researchers to use different tools. Extremely complex specifications can lead to model fitting problems. Sometimes, fitting problems indicate that the model is misspecified, so these problems are themselves informative. Hard to fit models may need to be simplified, especially if there are many small groups. Again, many small groups will also reduce the regularization benefits of the hierarchical model, as there is less information to estimate σ_θ .

Furthermore, hierarchical models can show general trends, but will not make powerful comparisons between every group treatment effect. Researchers who want to compare specific effects will often lack empirical leverage. This downside can also apply to other methods, however.

With these considerations in mind, when should researchers use hierarchical models in place of interactions? If only one factor modifies an effect, interactions are best, as the extra information hierarchical models provide is less valuable. Researchers should still remember power concerns with interactions, however.

With two or more modifiers, hierarchical models begin to add value because complex groups benefit from the conservative nature of partial pooling. Interpreting triple interactions between a variable and two modifiers is challenging. Fitting multiple pairwise interaction models, another common solution, raises multiple testing issues and does not account for correlations between modifiers. The advantages of hierarchical modeling increase with the number of modifiers until group sizes diminishes too much.

The best use cases of hierarchical models and machine learning are different. Unlike machine learning approaches, hierarchical models will not discover high-dimensional interactions. Researchers can add flexibility with additional interactions or non-linear specifications

in either level of the model, but this requires a priori specification. Therefore, if researchers want to focus on flexible discovery, not testing arguments with multiple sources of treatment heterogeneity, machine-learning is a better tool.

Data size is relevant to model selection as well. All of these models benefit from more data, as machine-learning models can draw out patterns and interactions have greater power. Hierarchical models also benefit from more data, but are less sensitive to outliers in small samples. The challenge is that some hierarchical models can take a long time to fit on large datasets. Variational approximation speeds model fitting but requires careful validation.

Table 1. *Key characteristics of different approaches to estimating heterogeneous effects.*

Characteristic	Hierarchical Models	Interactions	Machine Learning
Factors	Two or more	One or two	Many
Ideal Sample Size	Medium	Medium to large	Large
Complexity	Medium	Low	High
Computational Cost	Model dependent	Low	Variable
Interpretability	High	High	Low
Modifiers	Specified	Specified	Discovered/specified

Table 1 summarizes some relevant characteristics of hierarchical, interaction and machine learning approaches to heterogeneous effects. Interactions work well with one or two modifiers, a medium to large sample, and are easy to integrate with theory. Machine-learning tools are best for discovery with many modifiers in large datasets, but can be harder to connect with existing arguments. Hierarchical models are marginally more difficult to use than interactions, but researchers gain an interpretable way to tackle multiple modifiers in a theoretically informed model. I now document gains of hierarchical models compared to interactions with simulated data. I do not include machine-learning tools in the simulation because they often produce different effects, such as individual-level estimates that are not directly comparable to

the group-level hierarchical and interaction estimates.

3 Performance on Simulated Data

To assess how this hierarchical model compares to interactions, I first assess their performance on simulated data. Each simulation approximates the two most common applications of these models. The first simulates estimating treatment heterogeneity in factorial experiments that have a balanced probability of assignment to each group. The second simulation deals with imbalanced groups, as would likely be the case in demographic analyses of heterogeneous effects. In both simulations, I attempt to push the limits of group construction to illustrate when hierarchical modeling breaks down. A third simulation adjusts the data-generating process of heterogeneous effects to clarify the trade offs around partial pooling and priors in the presence of true outlier effects.

Both simulations fix the number of observations at 2,000 and manipulate the number of groups. In each simulation, the data generating process includes group-specific effects, drawn from a normal distribution with a mean of .2 and variance of .3. I also add three control variables with fixed coefficients that predict the mean of the simulated outcome. The simulated outcome has a standard deviation of one.

To compare the models, I use the root-mean squared error of the coefficient estimates compared to the true value. Root mean squared error is the best metric because the hierarchical model's shrinkage will introduce bias but compensate with lower variance (Clifford and Rainey, 2024). Lower variance may overcome greater bias in specific coefficients and make the hierarchical model more accurate on average. Pooling to the treatment mean is conservative in some ways, but that conservative approach is useful when we never know the true distribution of treatment effects. RMSE also captures instances where an estimator exaggerates the magnitude of the coefficient (Gelman and Carlin, 2014). Errors of sign and magnitude are just

as important as whether an uncertainty interval includes zero.

In the first simulation, I divide the 2,000 observations into groups based on combinations of three to seven binary variables, with equal probability of assigning observations to a group. This creates 8 to 128 unique groups, with approximately 250 observations per group at the low end and 16 observations at the high end. I then fit OLS models with fully crossed interactions of the group variables, as these estimate treatment effects in each subgroup. Fully crossed interactions are also a standard tool for estimating effects in each combination of experimental treatments. In the hierarchical model, I use main effects of the group variables in the heterogeneity equation and set a random intercept for each group. Fully crossing the group variables would saturate the model and make the random intercepts redundant.

Figure 2 presents the results of this first simulation. When the groups are large given three grouping variables, there is a small gap in model performance. The hierarchical model is marginally more accurate, however. But as group numbers rise and the number of observations in each group falls, root mean squared error rises for all models, but the OLS with fully crossed interactions performs much worse. With 128 groups and 16 observations per group, the OLS model has a root mean squared error of .5, while the hierarchical model is at .33.

These benefits are due to strongly reduced variance in the hierarchical estimates, as Figure 3 shows. This figure plots the true group treatment effects against the model estimates. Points above the dotted line are over-estimated by the model compared to the true value, while points below are under-estimated.

Even with 8 groups and 250 observations per group, more OLS estimates are unbiased, but two groups are clearly off. This means that even with slight bias for all groups, the hierarchical estimates have lower overall root mean squared error. As the number of groups increases and group size falls, the estimates for both models become more noisy, but the hierarchical model estimates are far less variable. Lower variance makes the hierarchical model much more accurate on average.

Equal Group Size: RMSE of Group Treatment Effect Estimates

n = 2000

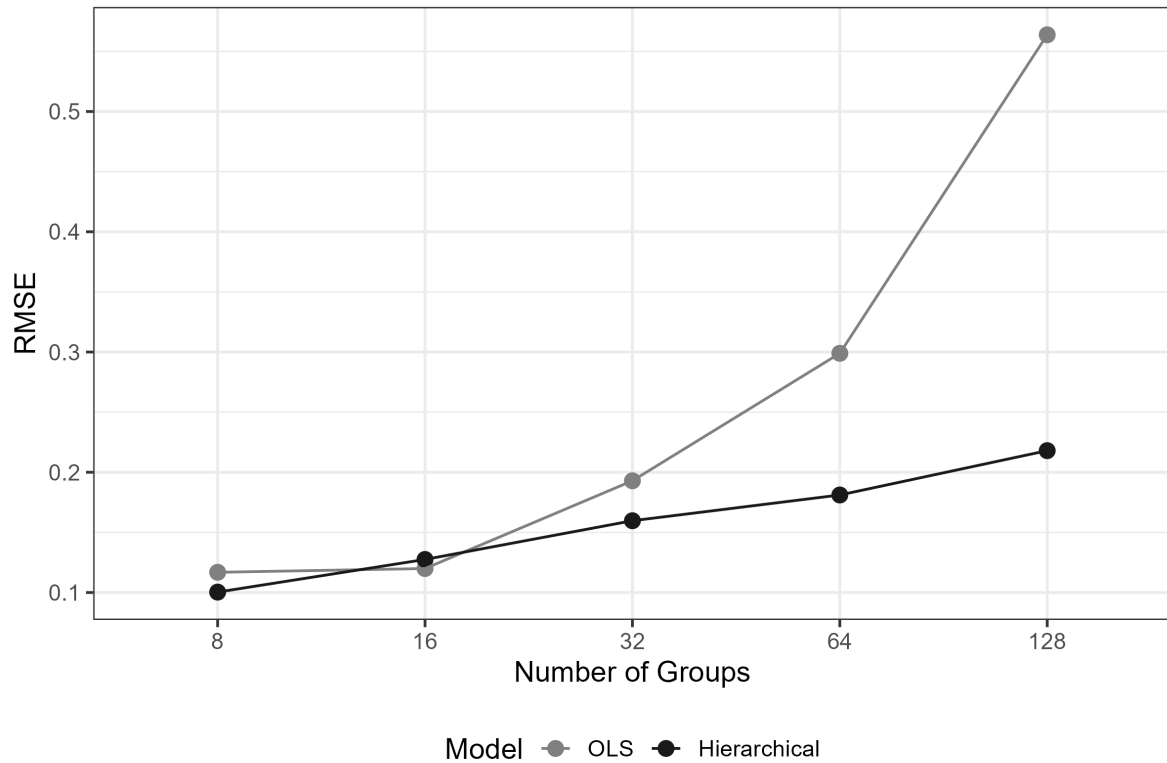


Figure 2. Comparison of OLS and hierarchical root mean squared error in group coefficient estimates. Simulation fixes the sample size at 2,000 observations and varies the number of groups. Each group has equal assignment probability.

Equal Group Size: Estimated vs. True Effects

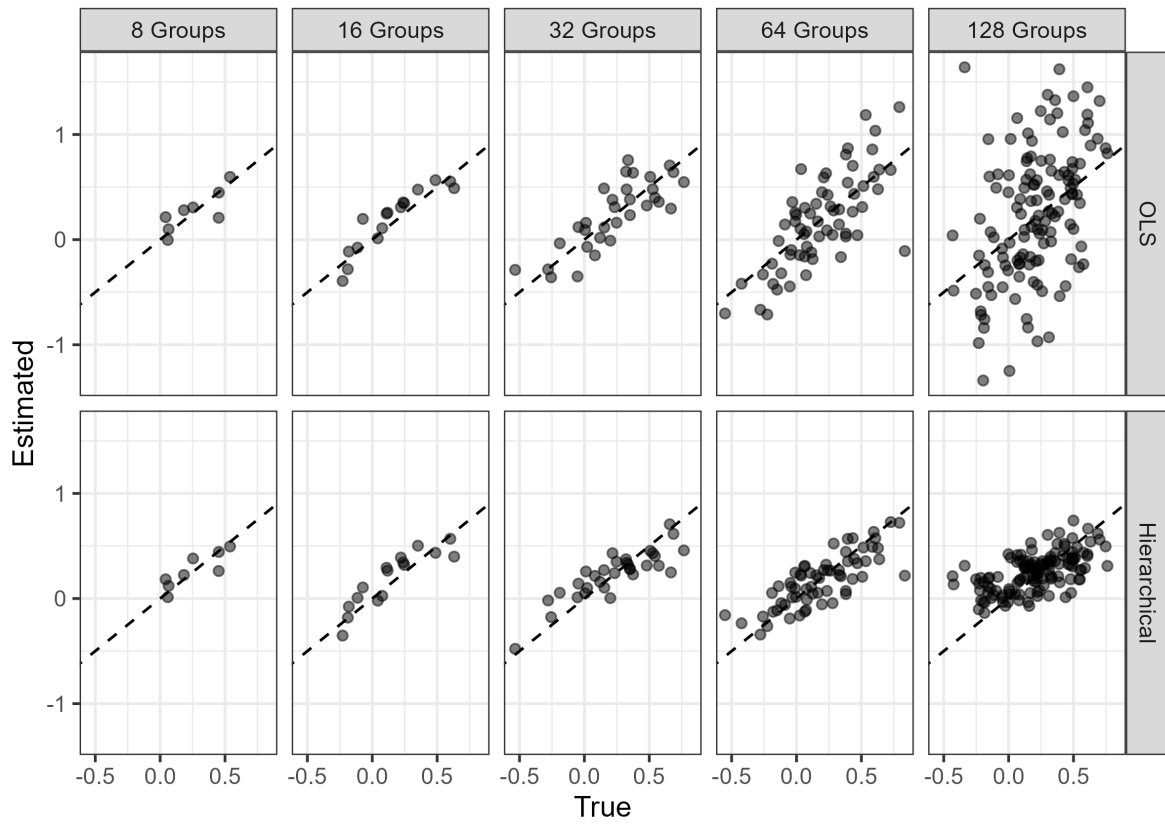


Figure 3. Comparison of OLS and hierarchical estimates of group-specific treatment effects. Points on the dashed line are unbiased.

This is essential because the dramatic over and under-estimates in the OLS model are the subgroup estimates with the greatest likelihood of statistical significance. But most of these estimates are very far from the true values and reflect noise in the data, not a true effect. The problems this poses for replication are now well understood, and the hierarchical model offers a clear solution.

The second simulation shows similar improvements with unbalanced group sizes. The relative benefits of hierarchical modeling fall as the number of small groups in the data rises, however. Here, I set the number of groups to 64, using six binary grouping variables. I then vary the probability of each group level and the symmetry of these probabilities across groups. Balanced groups are equivalent to the 64 group case in the first simulation. At the strongest imbalance, one group has a 20% share of ones and 80% zeros, and another has the opposite extreme, with a range in between. Such extreme imbalance can leave as few as one observation in each group, so it is an extreme test of model performance. Again, I compare the models with the root mean squared error of the coefficient estimates, and show these results in Figure 4.

Under balanced groups, the hierarchical model has a significant advantage. The smaller and more numerous the groups, the less valuable hierarchical regularization becomes, however. Small groups add far less information and are strongly pulled to the overall mean, which itself is a function of the larger groups. Again, the strong imbalance creates conditions where it is wiser to set up less finely-tuned groups or expand the sample size so groups have more observations and information to contribute to partial pooling.

To further illustrate how group size shapes the hierarchical estimates, I plot the bias of the OLS and hierarchical models in Figure 5. Again, the OLS estimator has significant bias for some groups as the model picks up noise. The magnitude of the bias does not depend on group size. In fact, large groups often have more bias because the OLS estimates weight each group equally.

The hierarchical model has much less bias for larger groups. Smaller groups with less

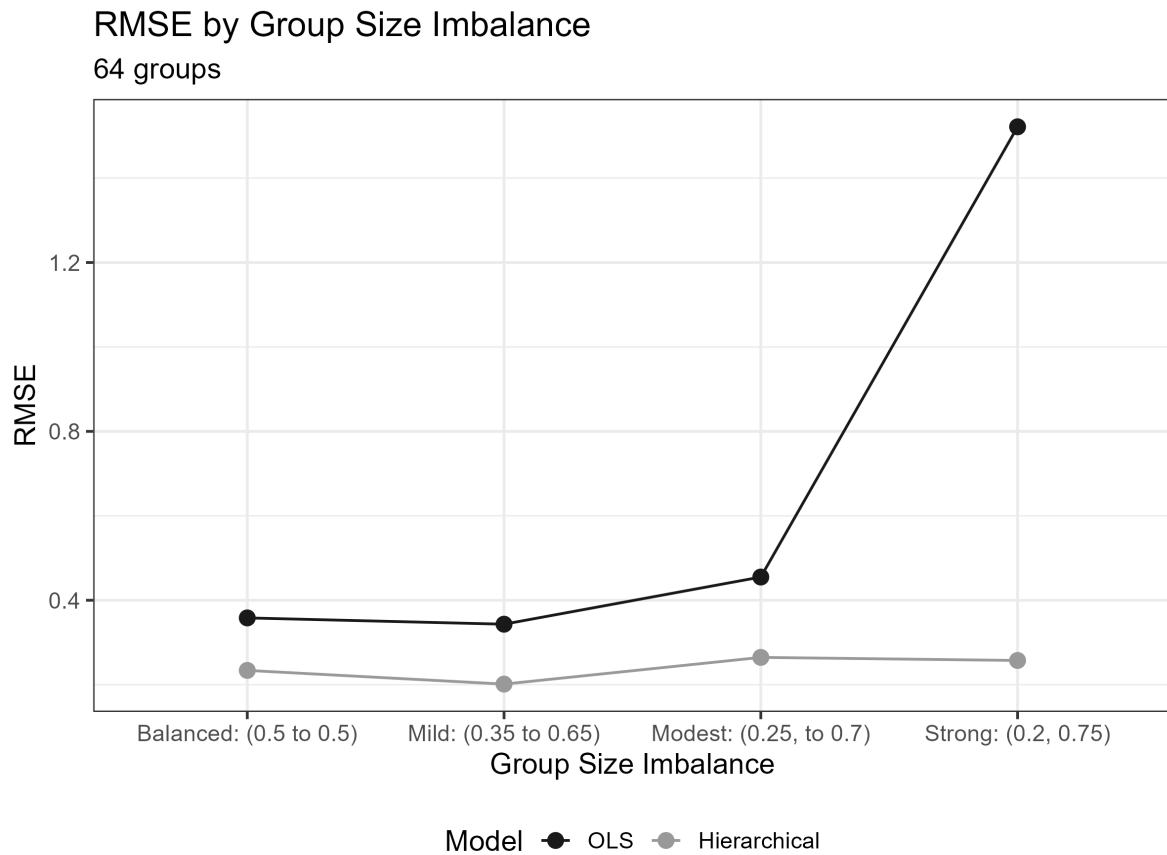


Figure 4. Comparison of OLS and hierarchical root mean squared error in group coefficient estimates. Simulation fixes the sample size at 2,000 observations and varies the relative size of different groups.

Absolute Bias by Group Size

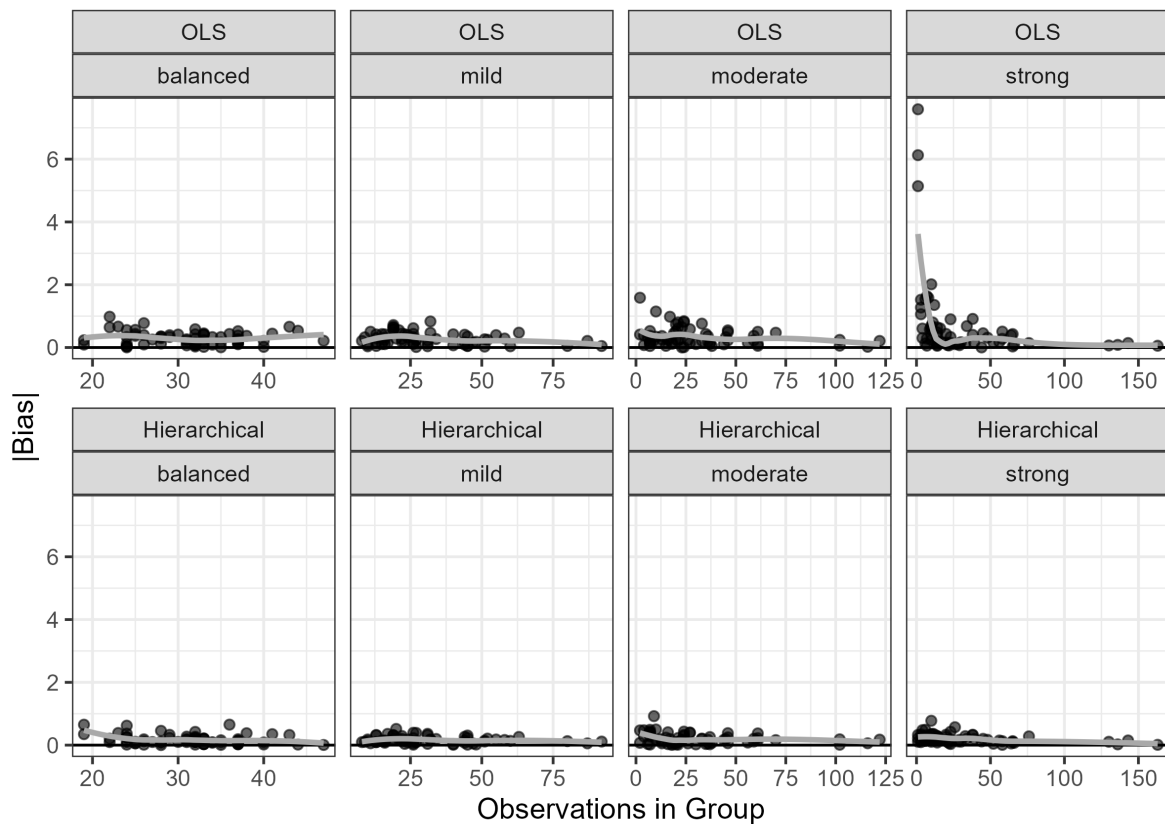


Figure 5. Comparison of OLS and hierarchical model bias in estimates of group-specific treatment effects. Group size gives the number of observations in each group, and bias is expressed as an absolute value. Loess lines give average bias across the range of observations.

than 20 data points have higher bias, while large groups, especially those with 75 or more observations, have more data and less bias. This reflects partial pooling. Small groups do not have much information, so the prior informs those estimates more than the limited data. Large groups have enough data to deviate from the prior and approach the true value.

Robustness of Hierarchical Model

In a third simulation, I examine the consequences of hierarchical models when the true distribution of effects includes significant outliers. In this simulation, I replace the normal distribution of effects with a student- t distribution with either five or three degrees of freedom. As the degrees of freedom in the student- t distribution drop, outlier effects become more common.

Because I still use a normal prior on the group effects, the hierarchical model has greater bias for extreme but true estimates. I document this pattern in Figure 6. While the hierarchical model still outperforms OLS in terms of overall accuracy and RMSE, hierarchical model bias increases for outlier effects. OLS has less bias for extreme values when effects come from a student- t distribution with three degrees of freedom, for example. This occurs because the prior pulls these large effects towards the overall mean.

Two things could mitigate this robustness concern. First, researchers should consider using half student- t priors on effect variance when they suspect potential outliers, and compare model fit with half-normal or half- t priors. Second, if outlier groups have much more data than other groups, they will be less sensitive to the prior, as the data overwhelms the prior variance hyperparameter. Small groups with true outlier effects will be especially biased towards the overall mean.

The simulations thus give two crucial inferences. First, partial pooling and modeling treatment effects offers significant improvements in inference, as it reduces variance in the estimates. The relative size of groups shapes the magnitude of this benefit, as hierarchical models will be

Student-T vs Normal Effect Distributions

32 groups, n = 2000

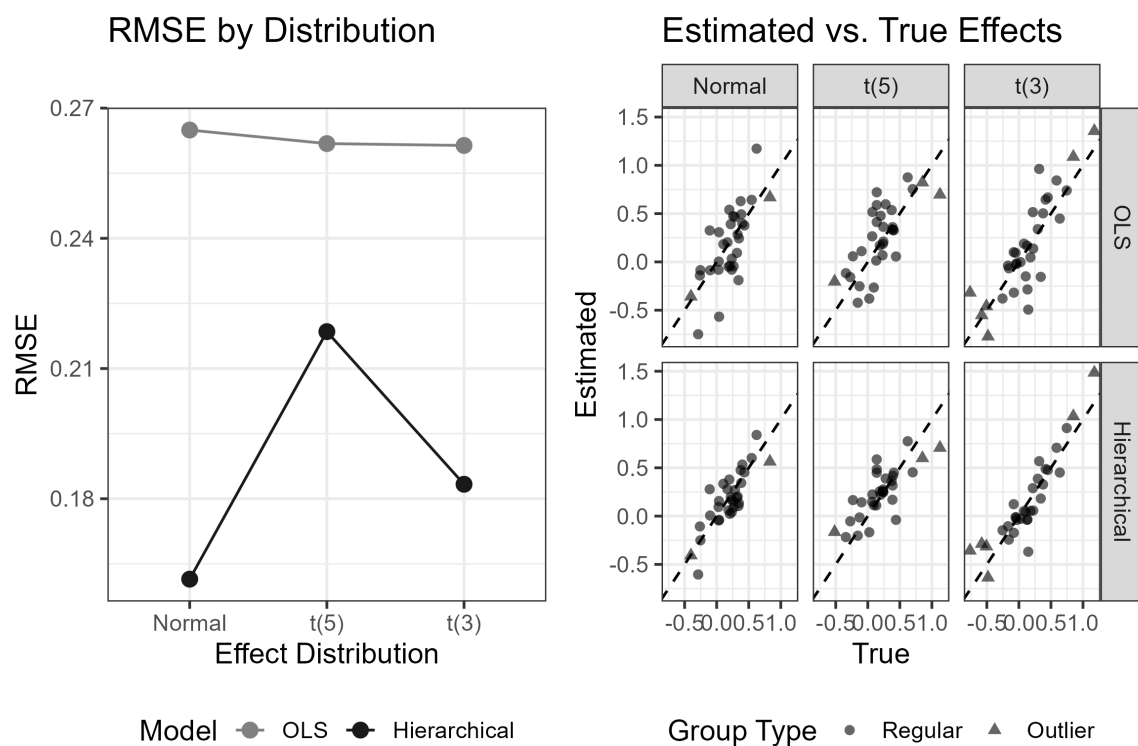


Figure 6. Comparison of OLS and hierarchical model estimates of group-specific treatment effects. The left panel gives the root mean squared error of the estimates, and the right shows true against estimated effects.

less biased for large groups and more biased with small groups under extreme imbalances in group size. This means that the smaller the group, the less a hierarchical model can tell you, but the estimate will often be more accurate than interactions regardless. Group estimates may not be perfectly unbiased and comparing groups will be difficult.

4 Example Application: Alliances and Public Support for War

In the following, I demonstrate how the hierarchical approach works in practice and compares to prior applied work reanalyzing a study by Tomz and Weeks (2021). Tomz and Weeks (TW hereafter) examine whether the public is more willing to go to war for when the beneficiary of that intervention is a U.S. ally. In a factorial experiment with vignettes, they find a 33% average increase in support for military intervention on behalf of allies, compared to non-allies. This is a large and potentially important relationship because the United States has a global network of allies.

Given the size of the main effect, TW's paper is an ideal scenario for comparing interactions and hierarchical models. Corresponding interaction effects may be large, and their sample size of 1,200 respondents is not unusual in published work. At the same time, TW estimated many interactions to check how other treatments modify the impact of alliances. There are 64 unique treatment groups with anywhere from 11 to 32 respondents, so estimates of the impact of alliances in the 32 pairs of alliance treatment and control groups employ at most 54 data points. As such, hierarchical regularization will likely change some conclusions by pooling noisy estimates in small groups. I document these gains by analyzing how other experimental treatments modify the impact of alliances, and then exploring how demographic differences modify the alliance treatment.

4.1 *Differences by Experimental Scenario*

Along with alliances, TW randomly assign whether the potential beneficiary of U.S. intervention is a democracy or not, the stakes of intervention, the potential costs, and the region of the world. They estimate the impact of alliances in the 32 treatment conditions with an OLS model that fully crosses interactions between the treatments, and calculate marginal effects over different averages of these groups. I use a hierarchical model to estimate the impact of alliances, with fully crossed experimental treatments as the systematic modifiers of the alliance effect and group-specific intercepts. I keep their Gaussian likelihood to minimize changes, although a logistic regression would fit the data better. The resulting approach mirrors TW's model, but regularizes the estimates with partial pooling.

Figure 7 compares the estimated alliance treatment effects across the experimental groups with the OLS and hierarchical models. This figure illustrates the regularization of hierarchical modeling in two ways. It shows the difference between the median hierarchical estimate and the OLS estimate for each combination of treatment groups and plots the distribution of effects.

First, the hierarchical estimates are much less variable and more tightly clustered around the overall mean. This occurs because the OLS model mechanically forces treatment effect estimates to the value of the coefficients and corresponding subgroup means. Given the size of this sample, the treatment effects are based on comparisons of roughly 25 treatment and 25 control respondents.

Second, the hierarchical estimates pull in unusual values. The bottom panel of Figure 7 shows the difference between the hierarchical and OLS estimate for each group. The downward slope from positive to negative values means that the hierarchical estimates are larger than below average OLS estimates, but smaller than above average interaction estimates. The hierarchical estimates shift towards the overall mean, away from extreme values. The OLS model finds one scenario where alliances reduce support by almost 20%, but the hierarchical model treats this as noise and brings it towards the grand mean, leading to a 50% difference.

Comparison of Heterogeneous Treatment Estimates

Heterogeneity from Experimental Conditions

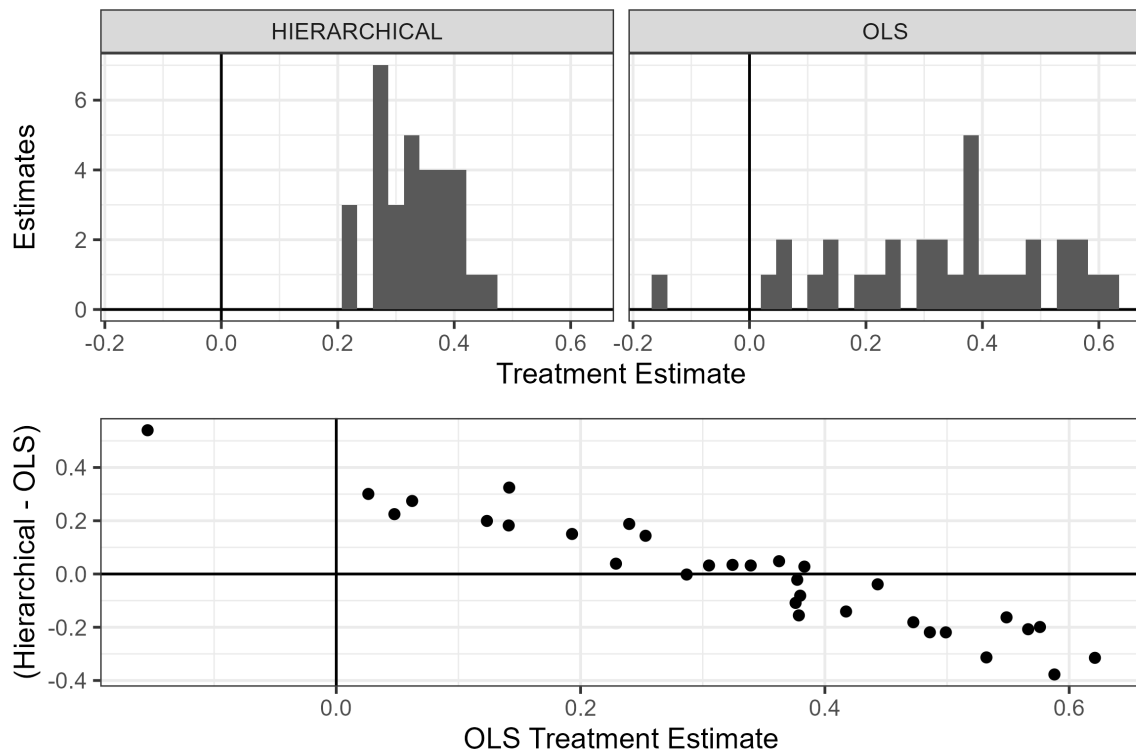


Figure 7. Comparison of OLS and hierarchical estimates of the impact of alliance across experimental conditions. The top panel gives a histogram of the treatment effects from each model. The bottom panel gives the difference between the hierarchical and OLS estimates for each group.

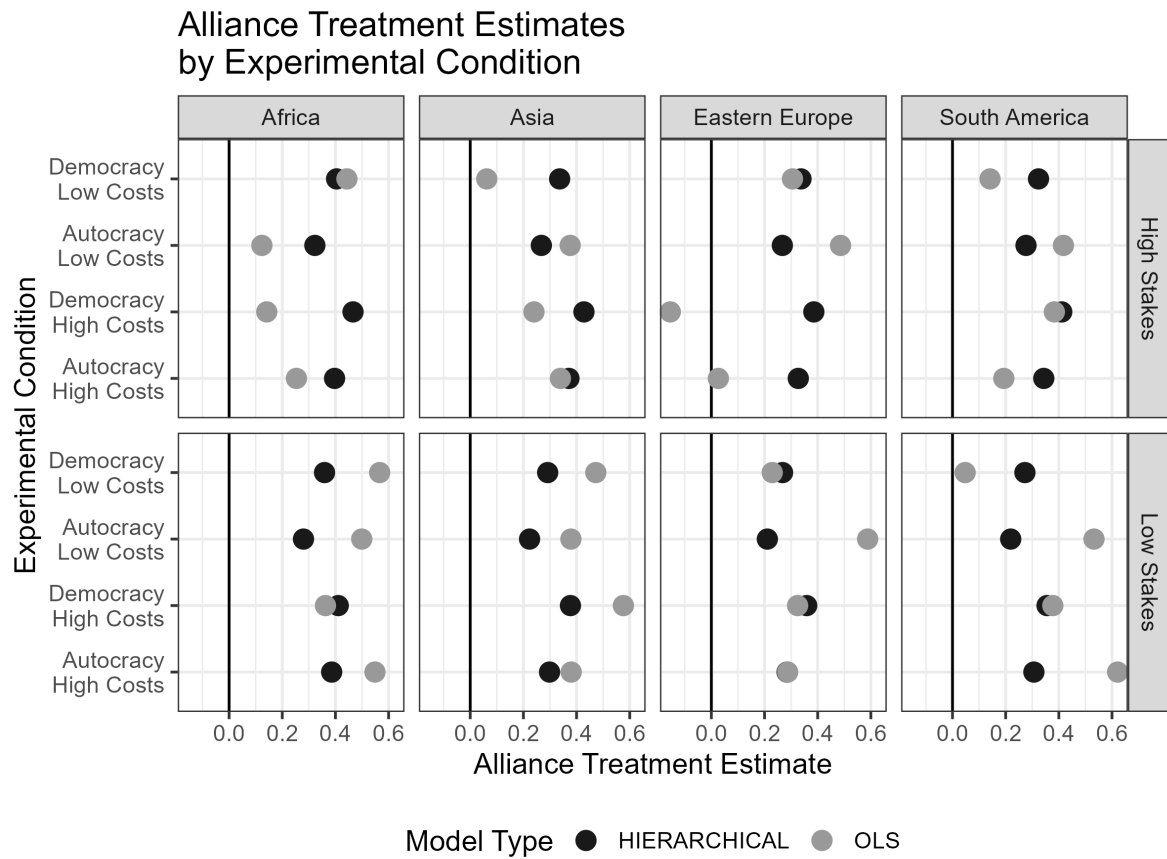


Figure 8. Comparison of OLS and hierarchical estimates of the impact of alliance across experimental conditions. Each point gives the effect estimate for each group.

I compare the estimates for each group with greater detail about what shapes each heterogeneous treatment in Figure 8. This clearly shows the reduced variation in the hierarchical estimates across the different experimental conditions. In some cases, the hierarchical and OLS estimates are very similar, but the model suggests that in a smaller sample, saturated interactions of treatment effects overstate effect heterogeneity. For some conditions, such as a low-stakes, low-cost intervention to support a South American autocracy, the hierarchical estimate is 30% lower than the OLS estimate. The condition where TW find a 20% drop in support for an ally compared to a non-ally is a high-cost intervention on behalf of an Eastern European democracy. That would be concerning, but the hierarchical model suggests that this is closer to a 30% increase. The 30% increase matches later findings on how randomly assigning NATO membership to different Eastern European states impacts U.S. public support for war (Tomz, Weeks and Bansak, 2023).

Given the very large average effect of alliances, these results do not change the direction of the findings- in some scenarios they even strengthen them. However, the magnitudes of some subgroup estimates do matter, as they might lead observers to misidentify when alliances matter most. The hierarchical models suggest that low stakes are not as influential as the OLS results indicate, for example. For other, smaller effects, regularization with a hierarchical model might change inferences in subgroup analysis.

4.2 Who Responds to Alliances

To further explore the potential application of hierarchical models, this section examines how demographic factors modify the impact of alliances. I used party, news interest, race, and gender to modify the impact of alliances. I selected these variables because Tomz and Weeks examine party and news interest as potential modifiers, but gender (Barnhart et al., 2020) and race are also salient modifiers. I control for other experimental manipulations, age and education. Age and education are important group-level controls because they are group-

level variables that are correlated across party, race and gender. Following TW's OLS analysis, I use a Gaussian likelihood, although the outcome is a binary variable.

The resulting hierarchical model thus encompasses four subgroup comparisons in TW's appendix while adding two more modifiers. All of the comparisons in the hierarchical model would require at least six pairwise interaction models. What TW present in four figures, I with two.

To start, I present the two regressions from the hierarchical model. Figure 9 plots how different variables shape either the impact of an alliance or support for intervention. One facet of this figure gives the coefficient estimates from the model of heterogeneity, and a second gives the control estimates.

High news interest is the strongest predictor of how alliances impact support for war. Respondents with high news interest respond to alliances by .15 more than others in this survey, and that relationship could be as small as .05 or as large as .22. This differs from TW's conclusion that respondents with high and low news interest respond in roughly the same way, likely because the hierarchical model accounts for other factors that are correlated with news interest.

Other demographic predictors do not differentiate responses to alliances as clearly. The impact of alliances may be somewhat greater for white and male respondents, but those differences are not clearly different from zero as the 95% credible intervals include zero and small negative values. Similarly, Republicans and Democrats are not all that different from independents in how they respond to alliances. This matches TW's conclusion that partisanship does not create substantial differences in alliance effects. The intercept in this equation is theoretically meaningful, as it captures the impact of alliances when all the modifiers are zero. For independent, non-white women with low news interest, alliances increase support for force by roughly .15.

The control equation suggests that high stakes, high costs, and partner democracy all in-

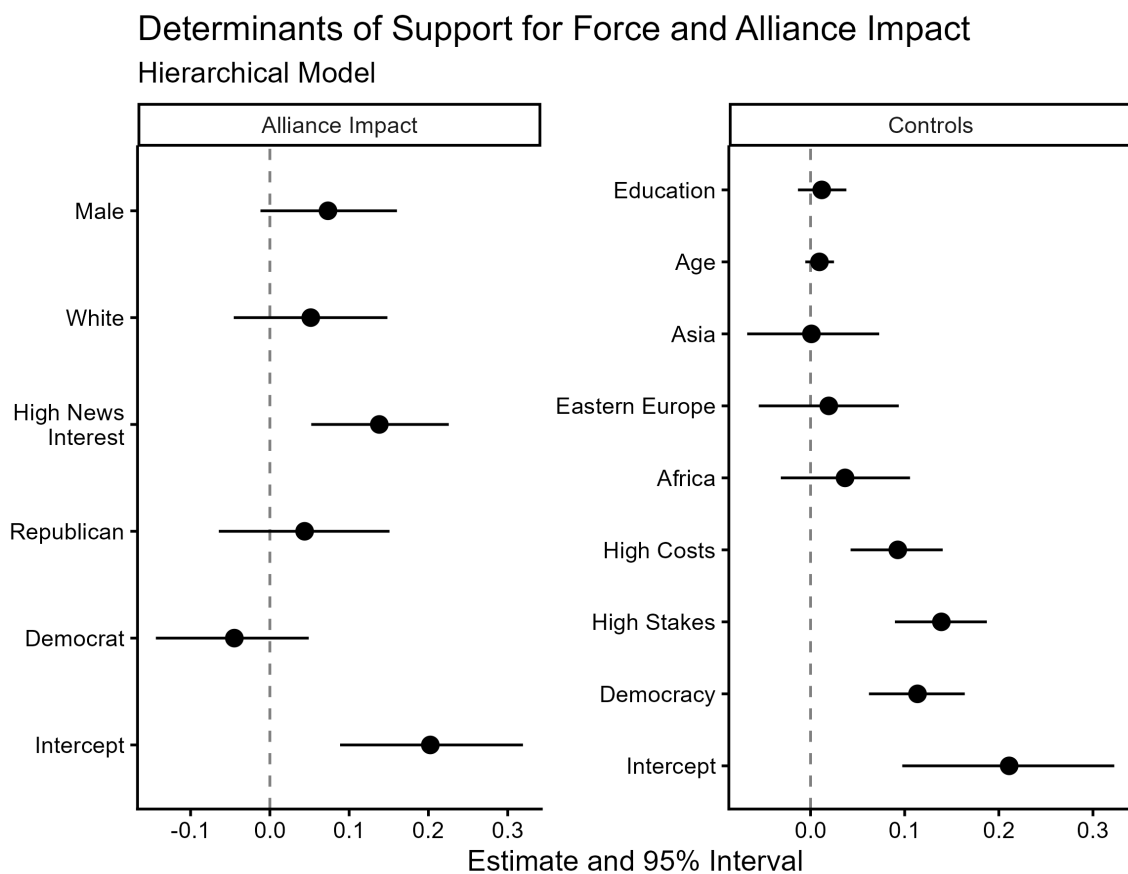


Figure 9. Regression coefficients from a hierarchical model of how an alliance shapes public support for military intervention. The left facet reports predictors of group alliance effects, and the right facet the control variables in the equation with the alliance treatment. Points give the posterior median and error bars the 95% credible interval.

crease support for intervention. Region is less consequential, as neither Asia, Eastern Europe nor Africa is clearly different from the reference category of Latin America. Because neither education nor age are equal to zero, the intercept is not directly interpretable.

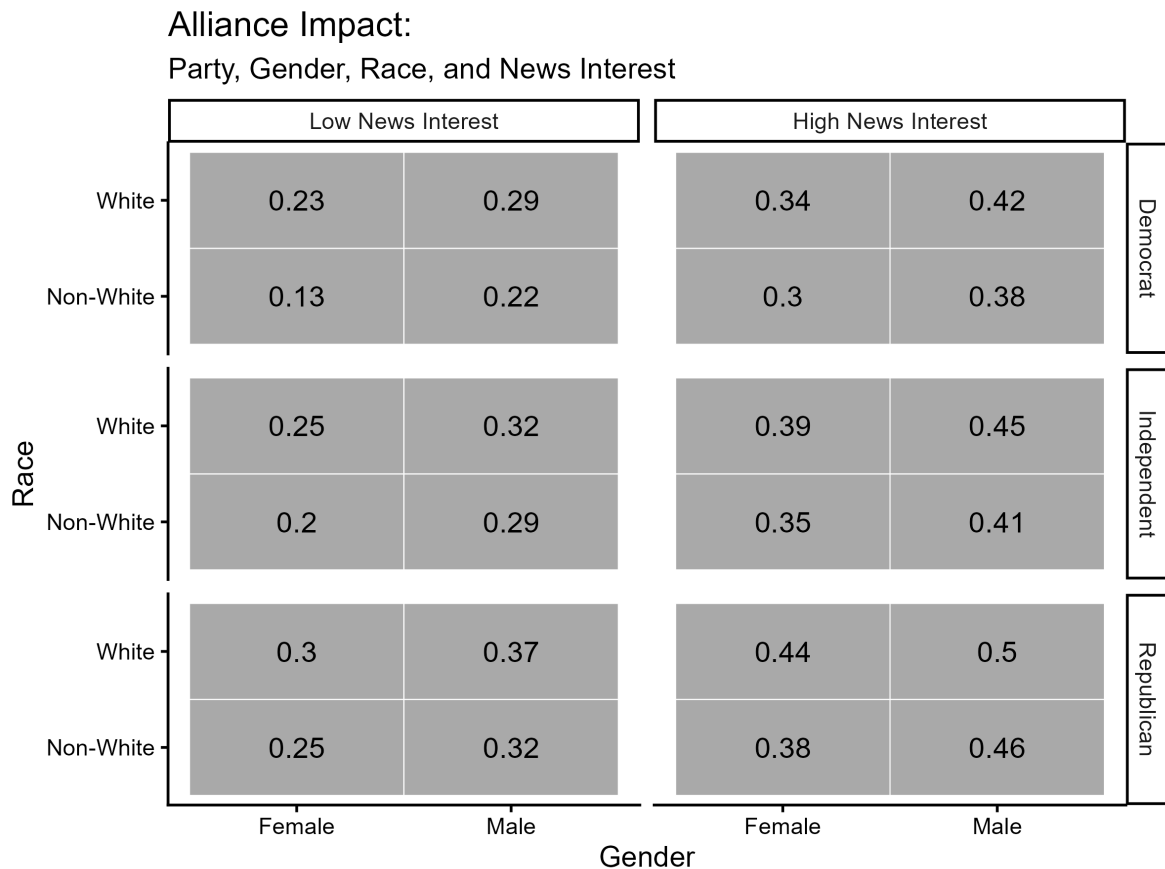


Figure 10. *Estimated impact of alliance on support for military intervention in different subgroups of respondents. Text gives the exact posterior median estimate.*

While it is possible to construct a rough profile of who responds most to alliances using the estimates in Figure 9, I provide a more precise summary in Figure 10. This figure plots the median estimate in each group of respondents, after grouping respondents based on race, gender, news interest and partisanship. There are 24 unique estimates, all of which are clearly greater than zero. Alliances thus increase support for using force among all demographic groups, but the magnitude of the effect varies.

Alliances most impact white Republican men with high interest in the news. In this group, the impact of alliances is .5, because this group was skeptical of intervention without an alliance rationale. The gap between high and low news interest in the other demographic groups is .15, which matches the coefficient in the heterogeneity equation.

The weakest impact of alliances appears for non-white, politically independent women with low news interest. Here, alliances increase support for force by 13%. White respondents respond by 8% more, Democrats by 4% more, and Republicans by 10%. The accumulation of these different effects create substantial gaps between the groups.

These results show some potential uses of the hierarchical approach to heterogeneous effects.⁹ Regularization moderates what might otherwise be extreme inferences about experimental subgroups. It also provides useful inferences about treatment heterogeneity that account for overlap among different sources of heterogeneity.

5 Conclusion

As conditional theories proliferate in social science, many independent variables have multiple potential modifiers. Interaction models cannot easily handle this shift, as adding three or more variables to an interaction reduces the interpretation benefits and fitting separate models for each modifier increases the risk of spurious results. Hierarchical modeling retains the clear interpretation of interactions in a single model that regularizes estimates.

I have therefore explained how to use hierarchical models to estimate heterogeneous effects. The first step first identify sources of heterogeneity, usually unit characteristics or other treatments, and set corresponding groups. Then, fit two regressions, one that models the relationship between a treatment and the outcome, and another that models the treatment effect with the modifiers and varying intercepts by group.

⁹In the appendix, I analyze Bush and Prather (2020).

Explicitly modeling how different groups respond to an independent variable can help test arguments and inform policy. It also details what drives variation in an effect and how much an effect varies. By using partial pooling of different groups, hierarchical modeling is less vulnerable to spurious inferences about subgroups.

The main downside of this model is that as a researcher adds more groups and group size shrinks, partial pooling becomes less useful. Small groups lack sufficient data to offset the prior, and are thus pooled strongly to the overall mean. If all the groups are small, this reduces the estimated variance in effect sizes so much that in extreme cases, it will collapse to basically zero. At this point, the systematic component in the regression will dominate and produce conclusions that are closer to interaction models.

Hierarchical modeling provides an intermediate approach between interactions and machine learning algorithms. For interactions with one or perhaps two modifiers, relying on simple interaction tools is best. Machine learning is best for discovery of complex heterogeneity, especially in large data. When there are two or more theoretically informed modifiers and researchers want to test their relative weight, hierarchical modeling allows flexible and interpretable estimation of effect variation.

Future work can build on this approach in other ways. Gaussian process regression is another flexible estimation tool with similar benefits that could extend to similar problems (Prati et al., 2023). Considering temporal, spatial and spatiotemporal autocorrelation in both equations is another worthwhile task. Another interesting application might be pooling experimental data and using the hierarchical structure to compare different manipulations in experiments with similar structure and smaller modifications. These tasks would further expand the benefits of hierarchical modeling in tackling multiple conditional arguments in a single framework.¹⁰

¹⁰AI Disclosure: I used Claude Code to refine simulation code and conduct the analysis of prior publications encapsulated in figure 1. I also checked the all the code for bugs or problems with Claude Code.

References

- Alley, Joshua. 2021. "Alliance Participation, Treaty Depth and Military Spending." *International Studies Quarterly* 65(4):929–943.
- Arel-Bundock, Vincent, Ryan C Briggs, Hristos Doucouliagos, Marco Mendoza Aviña and Tom D Stanley. 2025. "Quantitative Political Science Research Is Greatly Underpowered." *The Journal of Politics* . Available at: <https://osf.io/preprints/osf/7vy2f>.
- Barnhart, Joslyn N, Robert F Trager, Elizabeth N Saunders and Allan Dafoe. 2020. "The Suffragist Peace." *International Organization* 74(4):633–670.
- Blackwell, Matthew and Michael P Olson. 2022. "Reducing Model Misspecification and Bias in the Estimation of Interactions." *Political Analysis* 30(4):495–514.
- Brambor, Thomas, William Roberts Clark and Matt Golder. 2006. "Understanding interaction models: Improving empirical analyses." *Political Analysis* 14(1):63–82.
- Bürkner, Paul-Christian. 2017. "brms: An R package for Bayesian multilevel models using Stan." *Journal of Statistical Software* 80(1):1–28.
- Bush, Sarah Sunn and Lauren Prather. 2020. "Foreign Meddling and Mass Attitudes Toward International Economic Engagement." *International Organization* 74(2):584–609.
- Chaudoin, Stephen. 2014. "Promises or Policies? An Experimental Analysis of International Agreements and Audience Reactions." *International Organization* 68(1):235–256.
- Clark, William Roberts and Matt Golder. 2023. *Interaction Models: Specification and Interpretation*. Cambridge University Press.
- Clifford, Scott and Carlisle Rainey. 2023. Estimators for Topic-Sampling Designs. Technical report.
- Clifford, Scott and Carlisle Rainey. 2024. "Estimators for Topic-Sampling Designs." *Political Analysis* p. 1–14.
- Davies, Graeme AM and Robert Johns. 2013. "Audience costs among the British public: the impact of escalation, crisis type, and prime ministerial rhetoric." *International Studies Quarterly* 57(4):725–737.
- DiGiuseppe, Matthew and Paul Poast. 2018. "Arms versus Democratic Allies." *British Journal of Political Science* 48(4):981–1003.
- Dorie, Vincent, George Perrett, Jennifer L Hill and Benjamin Goodrich. 2022. "Stan and BART for Causal Inference: Estimating Heterogeneous Treatment Effects Using the Power of Stan and the Flexibility of Machine Learning." *Entropy* 24(12):1782.

- Feller, Avi and Andrew Gelman. 2015. "Hierarchical Models for Causal Effects." *Emerging Trends in the Social and Behavioral Sciences: An interdisciplinary, searchable, and linkable resource* pp. 1–16.
- Gelman, Andrew. 2008. "Scaling regression inputs by dividing by two standard deviations." *Statistics in medicine* 27(15):2865–2873.
- Gelman, Andrew. 2018. "You need 16 times the sample size to estimate an interaction than to estimate a main effect." Available at: <https://statmodeling.stat.columbia.edu/2018/03/15/need16/>.
- Gelman, Andrew and John Carlin. 2014. "Beyond Power Calculations: Assessing Type S (Sign) and Type M (Magnitude) Errors." *Perspectives on Psychological Science* 9(6):641–651.
- Goplerud, Max. 2021. "Modelling Heterogeneity Using Bayesian Structured Sparsity." *arXiv preprint arXiv:2103.15919*.
- Green, Donald P and Holger L Kern. 2012. "Modeling Heterogeneous Treatment Effects in Survey Experiments with Bayesian Additive Regression Trees." *Public Opinion Quarterly* 76(3):491–511.
- Grimmer, Justin, Solomon Messing and Sean J Westwood. 2017. "Estimating Heterogeneous Treatment Effects and the Effects of Heterogeneous Treatments with Ensemble Methods." *Political Analysis* 25(4):413–434.
- Guisinger, Alexandra and Elizabeth N. Saunders. 2017. "Mapping the Boundaries of Elite Cues: How Elites Shape Mass Opinion across International Issues." *International Studies Quarterly* 61(2):425–441.
- Imai, Kosuke and Marc Ratkovic. 2013. "Estimating treatment effect heterogeneity in randomized program evaluation." *The Annals of Applied Statistics* 7(1):443–470.
- Kertzer, Joshua D and Ryan Brutger. 2016. "Decomposing Audience Costs: Bringing the Audience Back into Audience Cost Theory." *American Journal of Political Science* 60(1):234–249.
- Künzel, Sören R, Jasjeet S Sekhon, Peter J Bickel and Bin Yu. 2019. "Metalearners for estimating heterogeneous treatment effects using machine learning." *Proceedings of the national academy of sciences* 116(10):4156–4165.
- Levendusky, Matthew S and Michael C Horowitz. 2012. "When Backing Down is the Right Decision: Partisanship, New Information, and Audience Costs." *The Journal of Politics* 74(2):323–338.
- Levy, Jack S, Michael K McKoy, Paul Poast and Geoffrey PR Wallace. 2015. "Backing Out or Backing in? Commitment and Consistency in Audience Costs Theory." *American Journal of Political Science* 59(4):988–1001.

- Lin-Greenberg, Erik. 2019. "Backing up, not backing down: Mitigating audience costs through policy substitution." *Journal of Peace Research* 56(4):559–574.
- Marquardt, Kyle L. 2022. "Language, Ethnicity, and Separatism: Survey Results from Two Post-Soviet Regions." *British Journal of Political Science* 52(4):1831–1851.
- Martin, Lisa L. 2005. "The president and international commitments: Treaties as signaling devices." *Presidential Studies Quarterly* 35(3):440–465.
- McElreath, Richard. 2016. *Statistical Rethinking: A Bayesian course with examples in R and Stan*. CRC Press.
- Prati, Annamaria, Yehu Chen, Jacob Montgomery and Roman Garnett. 2023. "A Gaussian Process Framework for Structured, Flexible, and Interpretable Machine-Learning Models."
- Ratkovic, Marc and Dustin Tingley. 2017. "Sparse estimation and uncertainty with application to subgroup analysis." *Political Analysis* 25(1):1–40.
- Schwartz, Joshua A and Christopher W Blair. 2020. "Do Women Make More Credible Threats? Gender Stereotypes, Audience Costs, and Crisis Bargaining." *International Organization* 74(4):872–895.
- Simmons, Joseph P, Leif D Nelson and Uri Simonsohn. 2011. "False-positive psychology: Undisclosed flexibility in data collection and analysis allows presenting anything as significant." *Psychological Science* 22(11):1359–1366.
- Stegmueller, Daniel. 2013. "How many countries for multilevel modeling? A comparison of frequentist and Bayesian approaches." *American Journal of Political Science* 57(3):748–761.
- Tomz, Michael. 2007. "Domestic Audience Costs in International Relations: An Experimental Approach." *International Organization* 61(4):821–40.
- Tomz, Michael and Jessica L.P. Weeks. 2021. "Military Alliances and Public Support for War." *International Studies Quarterly* 65(3):811–824.
- Tomz, Michael, Jessica LP Weeks and Kirk Bansak. 2023. "How membership in the North Atlantic Treaty Organization transforms public support for war." *PNAS Nexus* 2(7):pgad206.
- Wager, Stefan and Susan Athey. 2018. "Estimation and Inference of Heterogeneous Treatment Effects using Random Forests." *Journal of the American Statistical Association* 113(523):1228–1242.